GECG10069 (561084) F26: Introduction to Programming (C++)

Lab 2 : Literals & Variables



What you will learn from Lab 2

In this laboratory, you will further explore special output symbols and practice using variables in your programs.

TASK 2-1 : #DEFINE V.S. CONST NUMBER

- ✓ Figure out why coercion happened
- ✓ Try two different PI_1 & PI_2

```
#include <iostream>
using namespace std;

#define PI 3.14159
//#define PI_1 3.14159   // 巨集常數
//#define PI_2 0.14159 + 3
const double PI_1 = 3.14159;
const double PI_2 = 3 + 0.14159;

int main() {

    int r = 5;
    double area = PI * r * r;  // 自動型別轉換 (int → double)
    double area1 = PI_1 * r * r;  // 自動型別轉換 (int → double)
    double area2 = PI_2 * r * r;  // 自動型別轉換 (int → double)
    //double area2 = PI_2 * static_cast<double>(r) * r; // 強制轉型

    cout << "Circle radius = " << r << ", Area = " << area1 << endl;
    cout << "Area with explicit casting = " << area2 << endl;

    return 0;
}
```

EXERCISE 2-1: 2×2 MATRIX INVERSE

- ✓ You're asked to write a C++ program that calculates the inverse of a 2×2 matrix.
- ✓ Given a 2x2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- ✓ the determinant of A is:

$$\det(A) = ad - bc$$

- ✓ and the inverse matrix is:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Input

- ✓ Four matrix elements *a*, *b*, *c*, *d* would be given. Each element is an **integer** within the range of **-10 to 10**.
- ✓ You may assume that the determinant from given input is **non-zero**, so the given matrix is always invertible.
- ✓ In this assignment, you do **not** need to read input from the user. Instead, define the four matrix elements directly in your code as variables. TA would modify the matrix elements in the code during the demo.

Output

- ✓ Output the original matrix and its inverse matrix.
- ✓ Output format:

Original Matrix:

<a> <b>

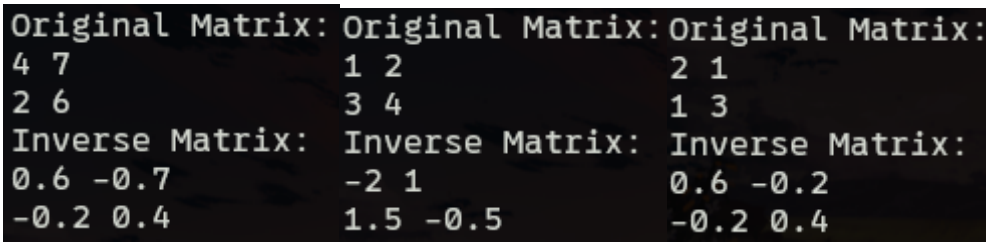
<c> <d>

Inverse Matrix:

<value> <value>

<value> <value>

Sample Output



Sample Code

```
#include <iostream>
using namespace std;


int main() {

    // Define variables, we will change matrix elements here (-10 ~ 10)
    int a = 4;
    int b = 7;
    int c = 2;
    int d = 6;

    // Compute the determinant


    // Compute the inverse matrix


    // Display


    return 0;
}
```

EXERCISE 2-2: VARIANCE

✓ You're asked to write a C++ program that calculates the average and variance of scores of three subjects: Math, Physics, Chemistry.

Formula

$$Average\ Score: \mu = \frac{Score_{Math} + Score_{Physics} + Score_{Chemistry}}{3}$$

$$Variance: \sigma^2 = \frac{(Score_{Math} - \mu)^2 + \left(Score_{Physics} - \mu\right)^2 + \left(Score_{Chemistry} - \mu\right)^2}{3}$$

Input

✓ Three subject scores would be given, each of which is an **integer** within the range of **0 to 100**.

✓ In this assignment, you do **not** need to read input from the user. Instead, define the scores directly in your code as variables. And TA would modify the scores in the code during the demo.

Output

✓ Output the score for each subjects, followed by the average score and variance, each on a separate line.

✓ Output format:

Math: <value>

Physics: <value>

Chemistry: <value>

Average: <value>

Variance: <value>

Sample Output

```
Math: 80          Math: 100         Math: 0
Physics: 90       Physics: 100      Physics: 0
Chemistry: 70     Chemistry: 100    Chemistry: 0
Average: 80       Average: 100      Average: 0
Variance: 66.6667 Variance: 0       Variance: 0
```

Sample Code

```
#include <iostream>
using namespace std;


int main() {

    // Define variables, we will change scores here (0 ~ 100)
    int math = 80;
    int physics = 90;
    int chemistry = 70;

    // Compute the average and variance


    // Display

    return 0;
}
```